ANGUS CHOU

(778) 668-0388 | tsuki9355@gmail.com | Portfolio | GitHub | LinkedIn

EDUCATION

Science Co-op | Computer Science Major @ University of British Columbia **Sept. 2025 - Expected Graduation in 2029**
Dean's List | GPA: 3.95 | Coursework: Data Structures & Algorithms, OOP, Software Engineering, Software Development

TECHNICAL SKILLS

Languages: Java, C#, C++, TypeScript, JavaScript, C, Verse
Frameworks & Tools: JUnit, Swing, React, Three.js, Unity, Unreal Engine 5, UEFN, WebGL, Git, GitHub, JSON
Engineering: OOP, DSA, unit testing, event-driven systems, finite-state machines, AI-assisted development, SQL

TECHNICAL PROJECTS

Vyxxen | *Unreal Engine 5, C++* | **GitHub**

- Developed a Zaxxon-inspired game in Unreal Engine 5 as part of Vertex School's Game Development Boot Camp.
- Exploring Unreal's Gameplay Ability System (GAS), delegates, event-driven architecture, and UObject lifecycle management to support scalable gameplay mechanics.
- Building familiarity with Unreal Engine's C++ gameplay framework, including custom classes, inheritance, composition, interfaces, and modular Actor Components to support reusable gameplay systems.

Pawn Escape | *Unity, C#* | **Play** | **GitHub**

- Built a turn-based game integrating BFS movement, FSM-driven turn system, and multi-system undo/rewind.
- Achieved **O(1)** tile occupancy lookups using dictionary-backed queries
- Replaced per-frame UI polling by designing an event-driven card interface.

Guess the Weight | *Unity, C#* | **Play** | **GitHub**

- Shipped a complete physics puzzle game solo in 48 hours. Enabled reusable level authoring and responsive interactions by building a physics-simulated balance scale, data-driven pipeline, event-driven item/UI systems, and custom editor tool.

Lumi the Forgotten Moon | *Unity, C#, Clip Studio Paint Pro* | **GitHub**

- Improved control responsiveness across movement states and transitions by implementing an FSM player controller with queue-based input buffering.
- Reduced gameplay coupling by separating movement, input, state logic, and behaviors into focused components using the Single Responsibility Principle.

CERTIFICATIONS & ADDITIONAL TRAINING

Vertex School | **AI Game Development Boot Camp** **Sept. 2026 - Nov. 2026**

- Developed 2 playable games using Unreal Engine 5 and C++, focusing on gameplay systems, modular architecture, and multiplayer functionality.
- Applying C++ object oriented programming concepts, including classes, inheritance, composition, interfaces, and reusable actor components, to structure gameplay logic.
- Exploring event-driven gameplay systems, delegates, UI/HUD state management, and AI-assisted development workflows with an emphasis on maintainable, reusable code.

EXPERIENCE

UBC Game Development Club | **Main Programmer** **September 2026 - Present**

- Collaboring with teammates as main programmer to design and implement gameplay systems for the UBC 2026 Winter game project.

AllSet Learning | **Math, English Tutor** **October 2025 - Present**

- Earned management recognition as a top-performing mathematics tutor by tutoring students and communicating detailed progress and learning needs.

Alphaville American Private School | **Teaching Assistant, File Manager** **July 2022 - July 2023**

- Supported school operations by formatting assignments, exams, report cards, and other Microsoft Word documents and preparing principal-facing classroom reports.